

Database Systems (CS2005)

Date: Monday, 15th Dec 2025

Time: 09:00 am to 12:00 pm

Course Instructor(s): Dr. Zulfiqar Ali Memon,
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Final Exam

Total Time (Hrs): 3

Total Marks: 50

Total Questions: 8

Do not write below this line

Attempt all the questions.

CLO # 2: *Analyze an information storage problem and derive an information model expressed in the form of an entity relation diagram and other optional analysis forms, such as a data dictionary.*

Q1: [marks: 5] [Estimated Time: 20 minutes]

Assume that immediate update recovery is being used in a DBMS.
A schedule at system crash is shown that includes five transactions.

Initial Values: A=5, B=35.

1. Design the log for the schedule in given Figure [2]
2. What are values after the crash for A and B. [2]
3. Specify which transactions need to undo? Provide justification for all the transactions, why do they need to undo or not. [1]

T1	T2	T3	T4	T5
R(A)				
	A = 10			
	W(A)			
		B = 10		
		W (B)		
		R(A)		
A = A + 10				
W(A)				
commit				
				B = 20
				W(B)
	commit			
		A = A + 10		
		W(A)		
		commit		
			R(A)	
			A = A + 15	
*****System Crash*****				

CLO # 2: *Analyze an information storage problem and derive an information model expressed in the form of an entity relation diagram and other optional analysis forms, such as a data dictionary.*

Q2: [marks: 4] [Estimated Time: 20 minutes]

Consider the following scenario below and answer the following parts:

An online food delivery platform handles thousands of simultaneous orders. Each order triggers parallel updates to inventory, rider assignment tables, and customer wallets. The system uses locking for consistency, but due to the heavy concurrency, the platform faces deadlocks several times per hour. Although the transactions are short, the deadlocks cause minor delays.

- a) Briefly explain one common Deadlock Prevention technique you would use. [2]
- b) Based on the scenario (high concurrency, small transactions, frequent deadlocks), should the database use Deadlock Prevention or Deadlock Detection & Recovery? Justify your answer by comparing the overhead and efficiency of both approaches in this environment. [2]

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Q3: [marks: 5] [Estimated Time: 20 minutes]

Part 1: Consider two transactions given below and explain why this blocking happens with binary locks and how S/X locks would improve the situation. **[2.5]**

- T1 reads item D and holds a binary lock.
- T2 wants to write D but keeps waiting because T1 is still reading.

Part 2: In below series of transactions, does T1 follow Strict 2PL, Basic 2PL, or none? Explain briefly. **[2.5]**

- T1: S-lock(A) → reads A → X-lock(B) → writes B
- T2: S-lock(A) → waits
- T1: unlock(A) → unlock(B)
- T2: S-lock(A) granted → reads A

CLO # 1: Explain fundamental database concepts.

Q4: [marks: 6] [Estimated Time: 25 minutes]

Imagine a cafe management system where customers place food orders, update their table reservations, and process bills concurrently. During a busy lunch hour, multiple transactions occur at the same time:

- a) Customer A places an order for a sandwich and juice, but the system crashes before the order is fully saved. **[2]**
- b) Customer B orders the last available slice of cheesecake. The system deducts the cheesecake from inventory, but before this deduction is committed, Customer C views the dessert menu on the tablet. **[2]**
- c) Customer D pays the bill via credit card. The bank successfully processes the payment, but immediately afterward the cafe's billing system crashes, and the payment status is not updated in the café records. **[2]**

Explain how the **ACID properties** would handle the above scenarios and ensure reliable and correct cafe-management transaction processing.

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Q5: [marks: 10] [Estimated Time: 30 minutes]

Consider the following 10 schedules of two transactions each. For each schedule, show the proper working and classify if it is: **recoverable, non recoverable, cascading rollback, cascadeless, strict, or conflict serializable**. A schedule may have more than one property. R1 defines read of transaction 1, W1 defines write of transaction 1, A1 defines abort of transaction 1, and C1 defines commit of transaction 1. Same goes for transaction 2.

1. R1(X); R2(X); W1(X); C1; W2(X); C2
2. R2(Y); W2(Y); R1(Y); W1(Y); C1; C2
3. R1(X); W1(X); R2(Y); W2(Y); C2; C1
4. R2(X); R1(X); W2(X); A2; W1(X); C1
5. W1(X); R2(X); W2(X); C2; C1
6. R2(X); W2(X); R1(X); A2; A1

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7. W2(Y); R1(Y); W1(Y); C1; A2
8. R1(X); R2(X); W2(X); C2; W1(X); C1
9. W1(X); W2(X); R1(X); C1; C2
10. R2(X); W2(X); R1(Y); W1(Y); C2; C1

CLO # 3: Demonstrate an understanding of normalization theory to normalize the database and formulate, using SQL & relational algebra, solutions to a broad range of query & data problems in a team work..

Q6: [marks: 10] [Estimated Time: 30 minutes]

Athletes, who are members of teams, compete in running events, which are held at fixtures throughout the year. primary keys are underlined and the foreign keys you have to identify yourself.

Athlete (AthleteID, Surname, Forename, DateOfBirth, Gender, TeamName)

EventType (EventTypeID, Gender, Distance, AgeGroup)

Fixture (FixtureID, FixtureDate, LocationName)

EventAtFixture (FixtureID, EventTypeID)

EventEntry (FixtureID, EventTypeID, AthleteID)

Using the above schema, Write down the SQL queries and relational algebra for the following questions

- 1) List all athletes with the events they entered, including fixture location
- 2) Count how many events each athlete entered
- 3) Count number of athletes in each event at each fixture
- 4) Show all teams and the number of athletes per team who entered events in 2018
- 5) Find events (distance + age group) that appear in more than 3 fixtures

CLO # 3: Demonstrate an understanding of normalization theory to normalize the database and formulate, using SQL & relational algebra, solutions to a broad range of query & data problems in a team work..

Q7: [marks: 5] [Estimated Time: 20 minutes]

Consider the following relation

Project_Grade (Project_id, Student_id, student_name, project_title, grade, submission_date, bonus)

The given FD's of above relation are:

FD1: student_id--> student_name

FD2: project_id-->project_title

FD3: submission_date--> bonus marks

Note: If any attribute is not mentioned in any FD, then by default they will be dependent on the whole primary key.

Identify partial and transitive dependencies (if any) and convert the table to 3NF and

- i) show all relation schema in 2NF. [2.5]
- ii) show all relation schema in 3NF. [2.5]

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Q8: [marks: 5] [Estimated Time: 25 minutes]

A super mart manages thousands of products, multiple departments, daily customer purchases, inventory tracking, suppliers, and billing operations. Customers can register with unique account IDs and maintain optional loyalty profiles that store reward points and purchase history. Products are organized into categories such as groceries, household items, fresh produce, dairy, and electronics. Each product has attributes like product ID, name, brand, price, category, and stock level. Suppliers, identified by unique supplier IDs, provide products to the super mart. Each supplier delivers multiple products, and each product may be supplied by more than one supplier. Supplier deliveries include delivery ID, date, quantity, and cost, contributing directly to inventory replenishment.

Customers purchase products through invoices/bills that contain invoice number, date, time, total amount, and payment method. Each invoice includes multiple products with details such as quantity, price per unit, and subtotal. Payments can be processed through cash, credit/debit cards, or mobile wallets, and each payment record contains payment ID, method, amount, timestamp, and status.

Inventory in each department (e.g., dairy, fruit & vegetable section, bakery, electronics aisle) is monitored in real time. Each department has a manager responsible for product arrangement and stock supervision. The system maintains stock ledger entries showing stock-in and stock-out records with timestamps.

Promotional campaigns and discount offers are applied to selected products, with details like offer ID, discount percentage, validity period, and applicable categories. The super mart ensures smooth functioning across procurement, inventory management, billing, payment processing, and customer service.

Your task is to create an Entity Relationship Diagram to show the complete flow of the platform. Also mention the structural constraints if required.

Good Luck!